Mechanisms for corrosion fatigue crack propagation

S. A. SHIPILOV
Department of Mechanical and Manufacturing Engineering, University of Calgary, Calgary, Alberta, Canada T2N1N4

Received in final form 20 July 2001

ABSTRACT The corrosion fatigue crack growth (FCG) behaviour, the effect of applied potential on corrosion FCG rates, and the fracture surfaces were studied for high-strength low-alloy steels, titanium alloys, and magnesium alloys. During investigation of the effect of applied potential on corrosion FCG rates, polarization was switched on for a time period in which it was possible to register the change in the crack growth rate corresponding to the open-circuit potential and to measure the crack growth rate under polarization. Due to the higher resolution of the crack extension measurement technique, the time rarely exceeded 300 s. This approach made possible the observation of a non-single mode effect of cathodic polarization on corrosion FCG rates. Cathodic polarization accelerated crack growth when the maximum stress intensity (K_{max}) exceeded a certain well-defined critical value characteristic for a given material-solution combination. When K_{max} was lower than the critical value, the same cathodic polarization, with all other conditions (specimen, solution, pH, loading frequency, stress ratio, temperature, etc.) being equal, retarded or had no influence on crack growth. The results and fractographic observations suggested that the acceleration in crack growth under cathodic polarization was due to hydrogen-induced cracking (HIC). Therefore, critical values of K_{max}, as well as the stress intensity range (ΔK) were regarded as corresponding to the onset of corrosion FCG according to the HIC mechanism and designated as K_{HIC} and ΔK_{HIC}. HIC was the main mechanism of corrosion FCG at $K_{max} > K_{HIC}$ ($\Delta K > \Delta K_{HIC}$). For most of the material-solution combinations investigated, stress-assisted dissolution played a dominant role in the corrosion fatigue crack propagation at $K_{max} < K_{HIC}$ ($\Delta K < \Delta K_{HIC}$).

Keywords corrosion fatigue; crack growth; fracture surface; high-strength steels; hydrogen-induced cracking; magnesium alloys; stress-assisted dissolution; titanium alloys.

INTRODUCTION

Over the past 30 years, the subject of corrosion fatigue crack growth (FCG) of metallic materials has attracted considerable attention from researchers throughout the world.[1] This stems from the fact that corrosion fatigue, along with stress corrosion cracking (SCC), is responsible for many, if not most, service failures in a wide variety of industries.[2] [Note: In fact, SCC is only a special case of corrosion FCG at the stress ratio (R = minimum/maximum load) equal to 1. SCC cannot be realized under realistic operating conditions because all real engineering structures are exposed to service stresses of varying amplitudes that usually are a mixture of stochastic and deterministic components.[3]] By now, the influence of mechanical, metallurgical, and environmental variables on the corrosion FCG behaviour of a range of materials has been studied in some detail. Nevertheless, despite the efforts which have been exerted, there is still no effective method for preventing corrosion FCG failures and it is not possible to predict which combinations of material and environment will result in corrosion FCG under in-service conditions.

The search for prevention methods, including the selection of materials resistant to the failure mode, is complicated by the fact that quite a number of fundamental questions regarding the possibility, mode and rate of corrosion FCG remain unanswered. Foremost among these questions is the problem of the corrosion FCG mechanism. The mechanisms, which have generally been proposed to explain corrosion FCG, are similar to

Correspondence: Sergei A. Shipilov, Department of Mechanical and Manufacturing Engineering, University of Calgary, Calgary, Alberta, Canada T2N 1 N4. E-mail: shipilov@enme.ucalgary.ca

those invoked in explaining SCC and began with research in SCC.[4] Two processes – hydrogen-induced cracking (HIC, or 'hydrogen embrittlement') and stress-assisted dissolution (SAD) of metal at the crack tip – are given main consideration as possible mechanisms of corrosion FCG. In the case of high-strength low-alloy steels and titanium alloys, for example, no firm consensus exists on the corrosion FCG mechanism in any one material-environment combination. For these materials, some authors propose that HIC plays a dominant role in corrosion fatigue crack propagation.[5–15] As evidence of HIC they point to the deleterious effect of increasingly negative cathodic potential on corrosion FCG rates[5–13] and the similarity between features of the fracture surfaces produced by crack growth in aqueous solutions and in hydrogen gas.[5,12,14,15] Others provide evidence that SAD is the dominant mechanism for corrosion FCG in high-strength steels[16–18] and titanium alloys.[19,20] In contrast to the emphasis placed on high-strength steels and titanium alloys, there is very little information on the corrosion FCG behaviour of materials such as magnesium alloys which have high strength-to-density ratios,[21] although much more effort has been devoted to studying SCC in magnesium alloys.[22–31] The mechanism of SCC of magnesium alloys has been ascribed to either continuous crack propagation as a result of SAD,[22,23,27] or discontinuous crack propagation as a result of a series of hydrogen-induced fractures at the crack tip.[26,28–31] Furthermore, SCC of magnesium alloys can involve the repeated cycles of a slow stage of SAD, followed by a fast stage of mechanical fracture.[24,25] One of the main arguments in favour of the SAD mechanism is that SCC of magnesium alloys can be stopped by cathodic polarization and restarted after polarization is removed.[22,27]

At least two techniques are preferred among those commonly used for identifying corrosion FCG mechanism(s). The first is the technique based on studying the effect of controlled (applied) potential on the corrosion FCG rates. Fractographic analysis of fracture surfaces is the second most popular technique and is usually employed in conjunction with the first technique. The basic idea of the controlled-potential technique was proposed by Uhlig in 1950[32] and then developed on by Phelps and Loginow in 1960[33] to distinguish between HIC and SAD as possible mechanisms of SCC in tests using smooth specimens. They suggested that, for example, if the application of small cathodic current decreases the time to failure, whereas small anodic current increases this time, the SCC mechanism operating in the absence of applied currents would be HIC. Similarly, the open-circuit mechanism is considered SAD if the time to failure increases with the application of small cathodic current and decreases with the application of small anodic current. Although at first glance this appears to be a simplistic approach, it has proved useful on occasions. In 1967, Leckie[34] adopted this approach to studying SCC in precracked specimens loaded statically. In 1971, Barsom[5] and Gallagher[6] pioneered the application of the controlled-potential technique to distinguish between HIC and SAD during corrosion fatigue crack propagation. In addition, Barsom,[5] Meyn,[14] and, more recently, Rungta and Begley[16] first examined in greater detail the features of the fracture surfaces produced by corrosion FCG under applied potentials by scanning electron microscopy (SEM) and transmission electron microscopy of plastic/carbon replicas. These examinations were conducted to provide further information concerning the corrosion FCG mechanisms. However, none of the above-mentioned attempts,[5,6,14,16] nor numerous attempts of other authors[7–13,35] who employed the controlled-potential technique and fractographic evidence, allowed for the possibility of quantifying the role of any one mechanism in corrosion fatigue crack propagation.

The present study is an extension of previous investigations[36–38] undertaken as a step towards the elaboration of a technique for identifying corrosion FCG mechanisms that could be applied to materials regardless of their composition and microstructure. Here, an attempt was made to quantify the role of HIC and SAD in corrosion fatigue crack propagation in metals that were of a different base, composition and microstructure. For this purpose, the well-known controlled-potential technique[5–14,16,35] was modified, and the effect of short-term electrochemical polarization on corrosion FCG rates of high-strength low-alloy steels, titanium alloys, and magnesium alloys was studied using views developed elsewhere to examine SCC mechanisms.[39,40] In addition, the fracture surfaces were examined by SEM to supplement the corrosion FCG data obtained under applied potential conditions.

EXPERIMENTAL DETAILS

Materials tested were martensitic steels 32Cr3NiMoV and 39Ni4MoV, titanium alloys VT20 (near-α) and TS6 (near-β), and magnesium alloys VMD10 and IMV7-1. Chemical compositions and mechanical properties of the materials are given in Tables 1–4. Fracture mechanics single-edge notched, pin-loaded specimens (210 mm long and 25 mm wide with an edge notch 3 mm deep) in the long transverse (LT) orientation were used for most experiments. The specimen thickness was 3 mm for steels and TS6 alloy, 8 mm for VT20 and IMV7-1 alloys, and 10 mm for VMD10 alloy. The central part of each specimen was ground and then the specimen was precracked to 6 mm (notch plus precrack). In addition, 15-mm thick compact tension specimens having an overall width of 62.5 mm, a net width (W)

Table 1 Chemical composition (wt%) of the high-strength steels

Steel	C	Si	Mn	P	S	Cr	Ni	Mo	V	W	Cu
32Cr3NiMoV	0.32	0.98	0.69	0.012	0.005	3.02	1.01	0.44	0.12	0.88	0.12
39Ni4MoV	0.39	0.09	0.10	0.007	0.005	0.20	4.50	0.34	0.09	–	0.18

Table 2 Chemical composition (wt%) of the titanium alloys

Alloy	Al	V	Mo	Zr	Cr	Fe	H	Ti
VT20	6.5	0.8	0.8	2.5	–	0.3	0.0052	Bal.
TS6	3.0	6.0	4.5	–	10.5	3.0	0.0060	Bal.

Table 3 Chemical composition (wt%) of the magnesium alloys

Alloy	Y	Zn	Gd	Zr	Cd	Mg
VMD10	7.6	1.7	–	0.3	1.5	Bal.
IMV7-1	5.3	–	4.9	0.3	–	Bal.

Table 4 Mechanical properties of the materials tested

Material	YS (MPa)	UTS (MPa)	El (%)	HRC	HRB*
32Cr3NiMoV	1600	1860	11	51	–
39Ni4MoV	1540	1870	12	49	–
VT20	870	1000	9	33	–
TS6	1540	1600	4	38	–
VMD10	240	320	13	–	30 (39)
IMV7-1	290	340	14	–	38 (65)

* HRB values after ageing at 220 °C for 24 h are given in parentheses.

Table 5 Material–solution combinations investigated

Materials	Solution composition*	pH
Titanium alloys	(0.1–1) M KOH	13.0–14.0
Magnesium alloys, steels, titanium alloys	0.1 M NaOH	13.0
Magnesium alloys, titanium alloys	(0.5–1) M NaOH	13.7–14.0
Magnesium alloys, steels	(0.02–0.05) M Na_2HPO_4	10.0
Steels	(0.01–0.05) M $Na_2B_4O_7$	9.2
Magnesium alloys, steels, titanium alloys	(0.01–0.5) M NaCl	7.0
Titanium alloys	0.6 M $FeCl_3$	1.3
Magnesium alloys	0.1 M H_2CrO_4	0.35
Magnesium alloys, steels, titanium alloys	(0.5–2) M H_2CrO_4	−0.15–0.15

* Moreover, NaCl was added to the (0.1–1) M KOH, (0.1–1) M NaOH and (0.1–2) M H_2CrO_4 solutions to yield NaCl concentrations of (0.01–0.1) M.

(measured from the loading line) of 50 mm, a total breadth (measured parallel to the loading line) of 60 mm were machined from a plate of VMD10 alloy in the short longitudinal (SL) orientation. Compact tension specimens were chevron-notched to an overall depth of 26 mm and had side grooves of 1.5 mm deep; they were precracked to provide the initial a/W (where a is the perpendicular distance from the loading line to the crack tip) ratio of 0.3. To minimize the current needed for maintaining a constant applied potential, the entire specimen surface that would be exposed to the test solution was coated with stopping-off lacquer, except for a narrow surface strip along the anticipated crack path.

Specimens were cycled under tension–tension loading with a symmetrical triangular load waveform at a cyclic frequency of 0.1 Hz. The stress ratio was kept constant between 0.1 and 0.9 during each individual test. In addition, a few tests were conducted under constant loading condition. Crack extension was monitored continuously by direct current electrical resistance measurements. With temperature stability no less than ± 0.5 °C, it was possible to detect as little as 0.001 mm crack extension.

Tests were performed at 25 $\pm$ 0.5 °C in several aqueous solutions which, except for (0.01–0.5) M NaCl, were specially chosen to allow for corrosion fatigue testing below K_{ISCC}. The compositions of the solutions are given in Table 5. Despite the fact that materials were apparently immune to static SCC in most of the solutions investigated, they were highly susceptible to static SCC in a standard 0.5 M NaCl solution.

Polarization of specimen was achieved by a potentiostat in conjunction with a Ag–AgCl reference electrode and a platinum counter electrode. Potentials reported in this work have been converted to the standard hydrogen electrode (SHE) scale. More complete experimental details and details of electrochemical measurement procedure have been previously described.[37,38]

During investigation of the effect of applied potential on corrosion FCG rates, polarization was applied after reaching an initial corrosion FCG rate $(da/dN)_i$ corresponding to the open-circuit potential (OCP) at a given maximum stress intensity (K_{max}) for a time period in which it was possible to register the change in the initial crack growth rate and to measure the crack growth rate $(da/dN)_p$ under polarization. Due to the higher resolution of the technique used to measure the crack extension, some change in $(da/dN)_i$ was observed about 10–150 s after polarization application. Time periods needed to measure $(da/dN)_p$ rarely exceeded 300 s. Therefore, polarization was removed as soon as $(da/dN)_p$ was measured and the system was returned to the potential equal to OCP. After that the experiment was repeated at the new level of K_{max}. The range of $(da/dN)_i$ was changed from about 5×10^{-9} to above 10^{-5} m cycle^{-1}. The degree of the polarization effect on the corrosion FCG was defined as the ratio of $(da/dN)_p$ to $(da/dN)_i$. It was deduced that, for most of the material–solution combinations that were tested, the crack-tip environment and the conditions of crack formation had not changed markedly during the 300 s because crack growth rates returned to initial values $(da/dN)_i$ after polarization was removed (exceptions will be given separately).

RESULTS

High-strength steels

On the basis of some preliminary tests, passivating solutions such as 0.05 M $Na_2B_4O_7$, 0.5 M and 1 M H_2CrO_4 were chosen to carry out most of the corrosion FCG tests and study the effect of applied potential on corrosion FCG rates in 32Cr3NiMoV and 39Ni4MoV steels. In these solutions, SCC due to SAD was suppressed as evident in the increase of K_{ISCC}. For example, $K_{ISCC} = 26$ and 35 MPa$\sqrt{m}$ for 32Cr3NiMoV steel tested in 1 M H_2CrO_4 and 0.05 M $Na_2B_4O_7$ solutions, respectively; whereas in 0.5 M NaCl solution $K_{ISCC} = 12$ MPa$\sqrt{m}$. This allowed corrosion FCG tests to be carried out under conditions of so-called true corrosion fatigue below K_{ISCC}.

When studying the effect of short-term cathodic polarization on corrosion FCG rates it was found that depending on the value of K_{max}, a given cathodic potential either increased (by several times over) or decreased corrosion FCG rates comparable to those at OCP. An important point was that cathodic potential accelerated crack growth when K_{max} and the corresponding corrosion FCG rate (da/dN) exceeded certain well-defined critical values. When K_{max} and da/dN were lower than the critical values, the same cathodic potential, with all other conditions (specimen, solution, pH, temperature, frequency, R, etc.) being equal, retarded or had no influence on crack growth rates. Cathodic polarization accelerated crack growth with some time delay after polarization was applied. Depending on the potential value, the delays were from 50 to 250 s. Delays decreased and $(da/dN)_p$ further increased as the applied potential became more negative. These delays seemed to reflect the kinetics for processes associated with reaching a new dynamic equilibrium within the growing crack as the potential was changed. When K_{max} and da/dN were lower than the critical values, cathodic polarization did not result in crack growth acceleration for 300 s but the prolongation of the polarization action for a time more than 300 s (e.g. up to 1500 s) resulted in steadily decreasing crack growth.

Figure 1 shows the diagram indicating the change in the corrosion FCG rates of 39Ni4MoV steel tested in 0.5 M H_2CrO_4 solution after cathodic polarization application when K_{max} was either lower or higher than 8.3 MPa$\sqrt{m}$. In the former case ($K_{max} = 8.1$ MPa$\sqrt{m}$), the crack was arrested when cathodic potential of -200 mV$_{SHE}$ (current density ~ 0.01 mA cm^{-2}) was applied but after about 210 s, the crack growth rate became equal to the initial value $(da/dN)_i$. In the second case ($K_{max} = 8.9$ MPa$\sqrt{m}$), the same cathodic potential increased the crack growth rate by a factor of 12.7 such that $(da/dN)_p = 1.2 \times 10^{-6}$ m cycle^{-1}. The crack growth acceleration began after a time delay of about 120 s after polarization was applied. After polarization was removed and the system was returned to the potential equal to OCP, the crack growth rate returned to the initial value $(da/dN)_i$. Figure 2(a) shows the relationship between the $(da/dN)_p/(da/dN)_i$ ratio and the initial crack growth rate $(da/dN)_i$ which was changed within the range from 2.5×10^{-8} to 4.5×10^{-6} m cycle^{-1}. As shown in Fig. 2(b), the accelerating effect of cathodic polarization on corrosion FCG rates for 39Ni4MoV steel tested in 0.5 M H_2CrO_4 solution at -200 mV$_{SHE}$ was more pronounced with higher R. For example, the higher ratios of $(da/dN)_p/(da/dN)_i$ for this material–solution combination were equal to 1.6 at $R = 0.1$ and more than 20 at $R = 0.85$.

The nature of the effect of cathodic polarization on corrosion FCG rates did not change when an acidic solution such as 0.5 M H_2CrO_4 (pH 0.15) was replaced by the neutral 0.01 M and 0.5 M NaCl (pH 7), weakly alkaline 0.01 M and 0.05 M $Na_2B_4O_7$ (pH 9.2) or more alkaline 0.03 M Na_2HPO_4 (pH 10) and 0.1 M NaOH (pH 13). For each of the steel–solution combinations tested, the ranges of K_{max} and da/dN where a given cathodic potential either increased or decreased corrosion FCG rates comparable to those at OCP were observed. In some instances, when the effect of short-term cathodic polarization on corrosion FCG rates was little or negligible,

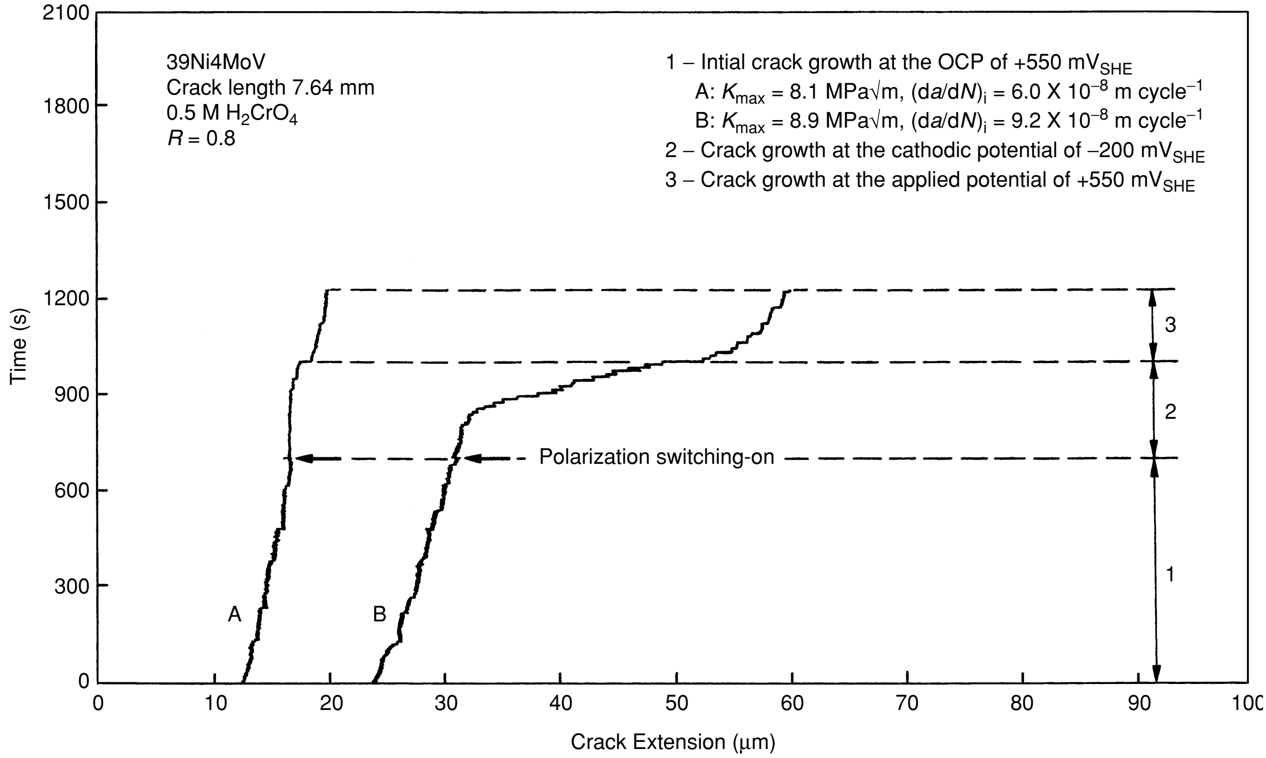

Fig. 1 Diagram fragments indicating the change in corrosion FCG rates in 39Ni4MoV steel after cathodic polarization application when K_{max} was either (A) lower or (B) higher than the critical value of 8.3 MPa$\sqrt{m}$.

it was concluded that polarization had no influence on the crack growth rate. For example, the cathodic potential of $-800\,mV_{SHE}$ (current density $\sim 0.1\,mA\,cm^{-2}$) had no visible influence on the crack growth rate in 32Cr3NiMoV steel exposed to 0.05 M $Na_2B_4O_7$ solution when K_{max} was lower than 14.7 MPa$\sqrt{m}$ at $R = 0.9$ but on reaching this critical value of K_{max}, the same cathodic potential increased the crack growth rate by a factor of 7.

Anodic polarization increased corrosion FCG rates over a wide range of $(da/dN)_i$ from 5×10^{-9} to above $10^{-5}\,m\,cycle^{-1}$. The accelerating effect of polarization became more evident as $(da/dN)_i$ was decreased. No critical values of K_{max} and da/dN such that the effect of anodic polarization on corrosion FCG rates changed sharply were observed.

Examination of fracture surfaces at low magnification ($10\times$) revealed no difference between fracture areas corresponding to crack growth under open circuit and accelerated crack growth under cathodic polarization when K_{max} exceeded critical values. The difference became evident at high magnification ($1250\times$) by SEM and apparently indicated that two different mechanisms of crack growth were operating under these conditions. For example, Fig. 3(a) shows a mixed intergranular (IG) and transgranular (TG) fracture mode of 32Cr3NiMoV steel tested in 0.05 M $Na_2B_4O_7$ solution at OCP. This mixed fracture mode changed to mostly IG at cathodic potential of $-1000\,mV_{SHE}$ as is shown in Fig. 3(b). The IG fracture mode occurred through the whole width of the specimen. An increase in the fraction of the IG fracture was accompanied by an increase in the amount of cleavage facets. Irrespective of potential, the fracture surfaces had fissures perpendicular to the fracture surface and main crack growth direction.

Titanium alloys

As was shown previously,[38] cathodic polarization retarded and even stopped corrosion FCG in VT20 and TS6 alloys, whereas anodic polarization slightly accelerated or had no influence on crack growth in most of the solutions investigated. Those were 0.5 M NaCl, (0.1–1) M NaOH, (0.1–1) M KOH, and (0.5–2) M H_2CrO_4 solutions. This polarization effect was observed over a range of K_{max} of 15 to 80 MPa$\sqrt{m}$ and $(da/dN)_i$ from about 10^{-8} to $5 \times 10^{-5}\,m\,cycle^{-1}$ and was consistent with published data.[19,35,41] Cathodic polarization retarded corrosion FCG rates irrespective of pH which varied from -0.15 for 2 M H_2CrO_4 solution to 14 for 1 M KOH solution. Figure 4 shows the results that were obtained in testing VT20 alloy in 0.5 M NaCl solution and 1 M KOH solution with and without NaCl additions.

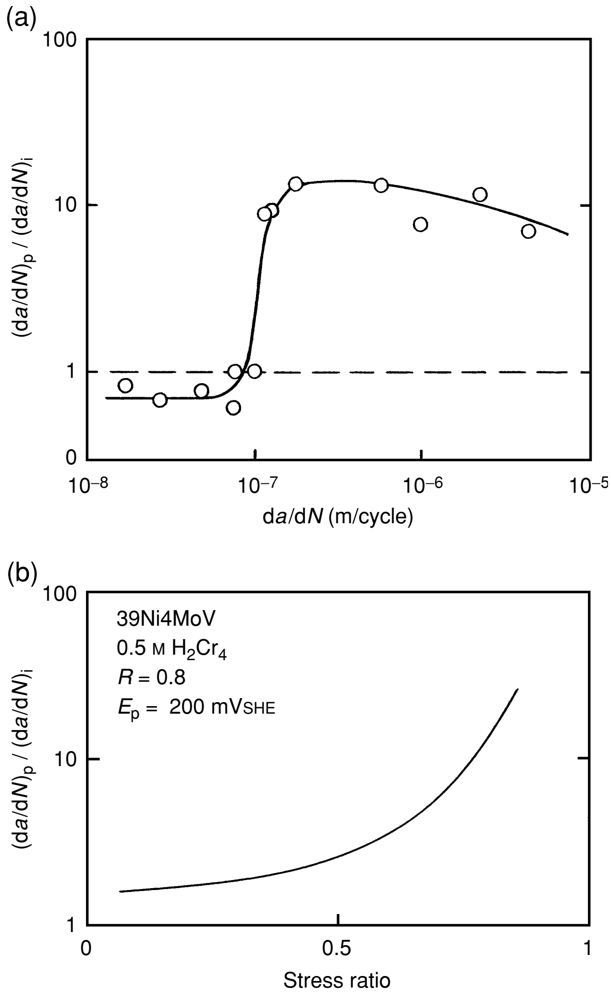

Fig. 2 Dependence of the $(da/dN)_p/(da/dN)_i$ ratio for 39Ni4MoV steel tested in 0.5 M H_2CrO_4 solution on: (a) the initial crack growth rate corresponding to OCP and (b) on stress ratio.

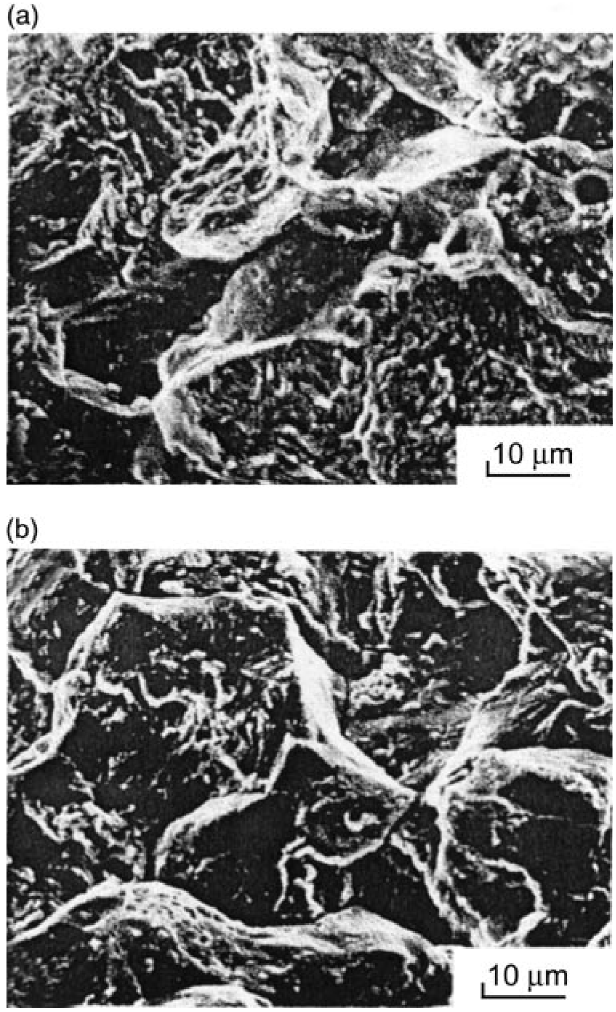

Fig. 3 SEM fractographs of 32Cr3NiMoV steel tested in 0.05 M $Na_2B_4O_7$ solution at: (a) OCP of $-50\,mV_{SHE}$ and (b) at cathodic potential of $-1000\,mV_{SHE}$; $K_{max} = 26\,MPa\sqrt{m}$, $R = 0.5, (da/dN)_i = 5.4 \times 10^{-7}\,m\,cycle^{-1}$, and $(da/dN)_p = 4 \times 10^{-6}\,m\,cycle^{-1}$. The overall direction of crack propagation was from top to bottom of the fractographs approximately.

Neither shifting the applied potential to more negative values nor decreasing the pH (down to 1) by addition of HCl resulted in accelerated crack growth in 0.5 M NaCl solution under cathodic polarization. Addition of a small amount of NaCl to hydroxide solutions resulted in their weaker passivating capabilities, and this manifested itself in some increase of crack growth rates. However, although the hydrogen penetrability of the passive oxide films formed on the crack-tip metal surface should increase due to weakening of the passivating capability of solutions,[42] cathodic polarization (up to potentials of 700–800 mV below the hydrogen equilibrium potential in the pH range tested,[43]) did not result in corrosion FCG acceleration in these solutions [Fig. 4(b)].

When testing the alloys in 0.6 M $FeCl_3$ and (0.5–2) M H_2CrO_4 + (0.01–0.1) M NaCl solutions, both a decrease and sharp increase in corrosion FCG rates under cathodic polarization were observed. Cathodic polarization accelerated crack growth by a dozen times comparable to that at OCP when K_{max} exceeded certain critical values. However, when K_{max} was lower than the critical values, the same cathodic polarization (with all other conditions being equal) retarded crack growth. As in the case of high-strength steels, cathodic polarization accelerated crack growth rates with some delay time after polarization was switched on. Figure 5 shows the diagram indicating the change in the corrosion FCG rate of TS6 alloy tested in 0.6 M $FeCl_3$ solution after the switching on of cathodic polarization when K_{max} was higher than the critical value of $65\,MPa\sqrt{m}$. Under these conditions, the cathodic potential of $-200\,mV_{SHE}$ (current density $\sim 10\,mA\,cm^{-2}$) increased the crack growth rate by a factor of 200 after

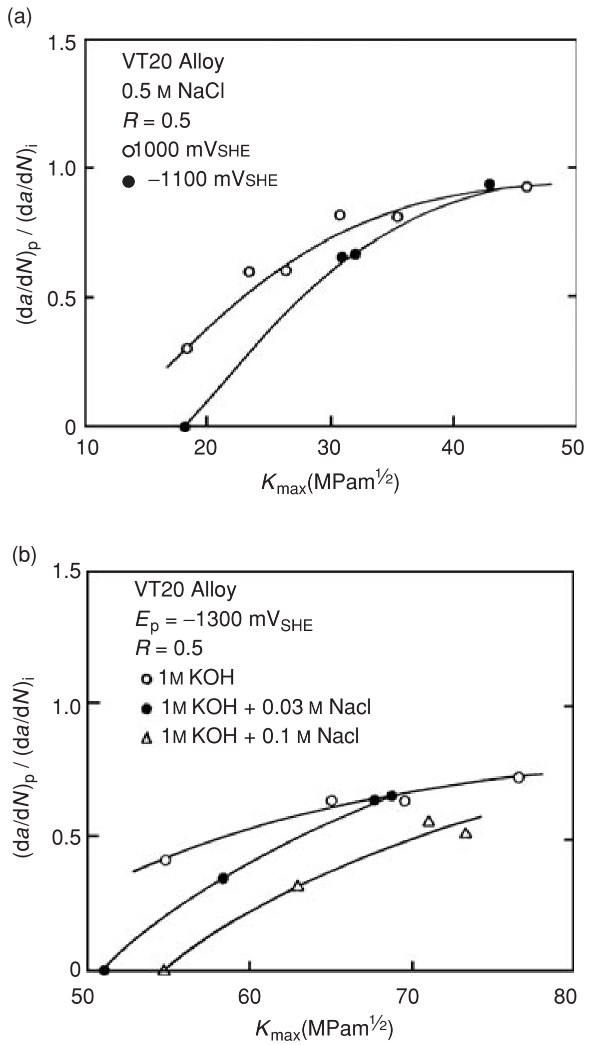

Fig. 4 Dependence of the $(da/dN)_p/(da/dN)_i$ ratio on K_{max} for VT20 alloy tested in: (a) 0.5 M NaCl solution, and (b) in 1 M KOH with and without NaCl additions.

about 130 s after the potential was applied. When testing VT20 alloy in 0.6 M $FeCl_3$ solution at cathodic potential of $-200\,mV_{SHE}$, these delays were 150 ± 15 s. Accelerated crack growth rates, when K_{max} exceeded the critical values, increased to a plateau level of 2×10^{-4} to 4×10^{-4} m cycle^{-1} where they were almost independent of K_{max}. (Because of the higher crack growth rates, the experiments were interrupted as soon as the accelerating effect of polarization became evident and the values of $(da/dN)_p$ needed to determine $(da/dN)_p/(da/dN)_i$ ratios were estimated from crack extension over three to five cycles). Immediately after polarization was switched off and the system was returned to the potential equal to OCP, crack growth rates significantly exceeded the initial value $(da/dN)_i$ for some time. When K_{max} was lower than the critical values, cathodic polarization did not result in crack growth acceleration during 300 s. On the contrary, polarization decreased the growth rate almost immediately after it was applied.

Figure 6(b) shows that cathodic polarization at $-200\,mV_{SHE}$ decreased corrosion FCG rates in VT20 and TS6 alloys tested in 0.6 M $FeCl_3$ at $R = 0.8$ when K_{max} values were lower than 63.0 and 72.6 $MPa\sqrt{m}$, respectively; the corresponding critical crack growth rates were 4.5×10^{-7} m cycle^{-1} for VT20 alloy and 6.8×10^{-7} m cycle^{-1} for TS6 alloy [Fig. 6(a)]. On reaching these values of K_{max}, the same cathodic polarization increased crack growth rates by a factor of 50 in VT20 alloy and by a factor of 70 in TS6 alloy. A strong dependence between $(da/dN)_p/(da/dN)_i$ ratios and R was observed. For example, for TS6 alloy tested in 0.6 M $FeCl_3$ at $-200\,mV_{SHE}$, the higher ratios of $(da/dN)_p/(da/dN)_i$ were equal to 5 at $R = 0.1$, 30 at $R = 0.5$ and 75 at $R = 0.8$. The critical value of K_{max} depended on the solution chemistry and the ratio of activating/passivating species. For example, for VT20 alloy tested at $-200\,mV_{SHE}$ and $R = 0.8$, the critical $K_{max} = 48.5\,MPa\sqrt{m}$ in 2 M $H_2CrO_4 + 0.1$ M NaCl solution, 43.5 $MPa\sqrt{m}$ in 1 M $H_2CrO_4 + 0.1$ M NaCl solution, and 27 $MPa\sqrt{m}$ in 0.5 M $H_2CrO_4 + 0.1$ M NaCl solution.

Examination of fracture surfaces, even with the naked eye, revealed the essential difference between fracture areas corresponding to crack growth under open circuit and accelerated crack growth under cathodic polarization when K_{max} exceeded a critical value. Crack growth under cathodic polarization was more brittle than under open circuit. This difference became more evident at higher magnification (640×) by SEM and testified to the change in the crack growth mechanism. Figure 7(a) shows that characteristic features of the fracture surface of VT20 alloy tested in 0.6 M $FeCl_3$ at OCP where well-defined fatigue striations perpendicular to the direction of main crack growth can be seen. On the other hand, many typical flat cleavage facets with characteristic 'river patterns' and flutes and fissures perpendicular to the fracture surface and virtually parallel to the main crack growth direction were evident when the alloy was tested at the cathodic potential of $-200\,mV_{SHE}$ [Fig. 7(b)].

Magnesium alloys

For studying the corrosion FCG behaviour of VMD10 and IMV7-1 alloys below K_{ISCC}, passivating solutions such as (0.1–1) M NaOH, (0.02–0.05) M Na_2HPO_4, and (0.1–2) M H_2CrO_4 were chosen. In accordance with Ref. [21], the given solutions attack magnesium and its alloys at a very low rate. Moreover, NaCl was added to the (0.1–1) M NaOH and (0.1–2) M H_2CrO_4 solutions to yield NaCl concentrations of (0.01–0.1) M. These mixed solutions, with the activity provided by the chloride and

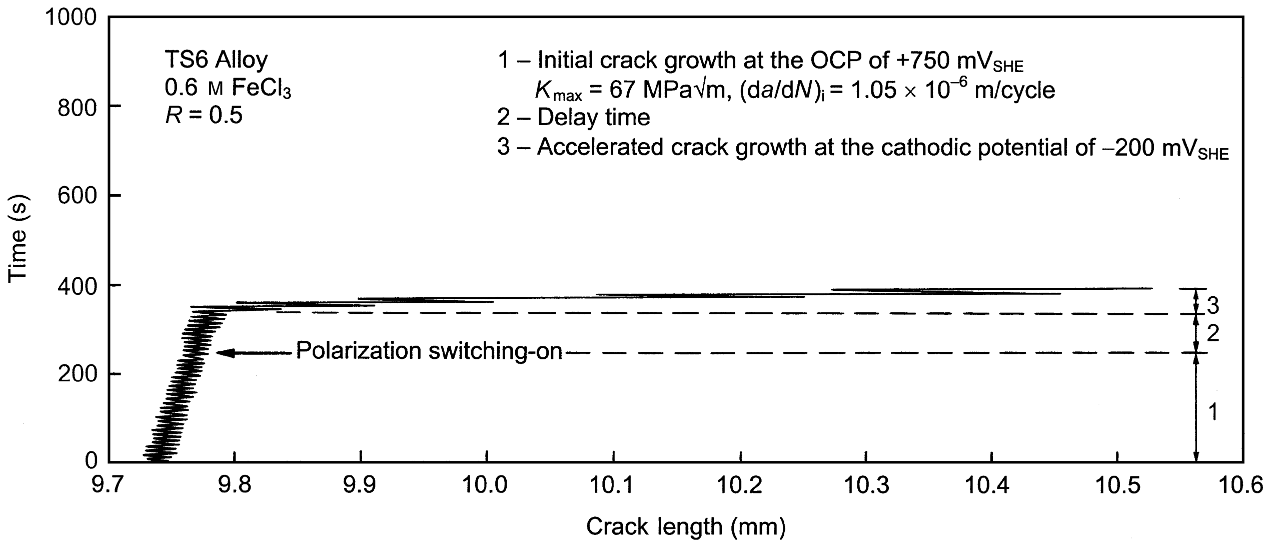

Fig. 5 Diagram fragment indicating the change in the corrosion FCG rate in TS6 alloy after cathodic polarization application when K_{max} was higher than the critical value of 65 MPa$\sqrt{m}$.

the passivating influence of the hydroxide or chromic acid, were expected to promote corrosion FCG and to increase the hydrogen permeability of passive films at the crack-tip metal surface. The latter condition was expected due to a weakening of the passivating capacity of solutions by adding chloride ions to the solutions.[44] In addition, pure (0.01–0.5) M NaCl solutions were used in the corrosion FCG tests.

Cathodic polarization retarded corrosion FCG in magnesium alloys exposed to (0.01–0.5) M NaCl solutions, whereas anodic polarization accelerated it. For IMV7-1 alloy, a similar effect of cathodic and anodic polarization was observed in all solutions investigated. An increase of the yield strength of both alloys by ageing at 220 °C for 24 h, carried out to increase their susceptibility to HIC, did not change the retarding effect of cathodic polarization on crack growth rates in NaCl solutions. The observed effect of polarization on corrosion FCG rates was consistent with that for SCC of magnesium alloys.[22,27,29,30]

When testing VMD10 alloy in (0.1–1) M NaOH, 0.02 M Na_2HPO_4, and (0.5–2) M H_2CrO_4 solutions, where FCG rates were approximately equivalent to those in air,[45] cathodic polarization accelerated corrosion FCG rates by several times comparable to those at OCP. This accelerated effect was observed only when K_{max} exceeded certain critical values. Depending on applied potential, the crack growth rate increased slowly in periods from 10 to 150 s after applying cathodic polarization and then the crack growth rate reached a steady-state value $(da/dN)_p$. When K_{max} was lower than the critical values, the same cathodic polarization had no obvious influence on crack growth rates over 300 s.

Figure 8 shows results obtained in testing VMD10 alloy (HRB 39, LT orientation) in 1 M NaOH solution at $R = 0.5$. Cathodic polarization at $-1800\ mV_{SHE}$ (current density $\leq 3\ mA\ cm^{-2}$) had no influence on crack growth when K_{max} and the corresponding FCG rate da/dN were lower than 13.2 MPa$\sqrt{m}$ and 5.6×10^{-7} m cycle^{-1}, respectively. However, on reaching these values of K_{max} and da/dN, the same cathodic potential accelerated crack growth in comparison to that at OCP. Polarization increased crack growth rate by a factor of 1.5 at $K_{max} = 13.6$ MPa$\sqrt{m}$ and the degree of the accelerating effect of polarization on crack growth rate increased to 6 at $K_{max} = 16.4$ MPa$\sqrt{m}$. A strong dependence between critical values of K_{max} and the specimen orientation was observed. In particular, when testing VMD10 alloy (HRB 39) in 1 M NaOH solution at $-1800\ mV_{SHE}$ and $R = 0.5$, the critical K_{max} was equal to 13.2 MPa$\sqrt{m}$ for LT orientation and only 4.9 MPa$\sqrt{m}$ for SL orientation. An increase of the yield strength of VMD10 alloy by ageing at 220 °C for 24 h was accompanied by a noticeable decrease of the critical K_{max}. For example, after ageing of SL specimens, the critical value of K_{max} decreased from 13.5 to 3.3 MPa$\sqrt{m}$ when testing VMD10 alloy in 2 M H_2CrO_4 solution at $-800\ mV_{SHE}$ and $R = 0.5$.

Addition of a small amount of NaCl to hydroxide and chromic acid solutions, as well as decreasing their concentrations to less than 0.1 M for NaOH and less than 0.5 M for H_2CrO_4, resulted in a weaker passivating capacity. It manifested itself in some increase of crack growth rates and the shift of da/dN data toward lower K_{max}. Thereafter, the effect of cathodic polarization on crack growth rates was reversed and cathodic polarization always retarded crack growth independently of K_{max}.

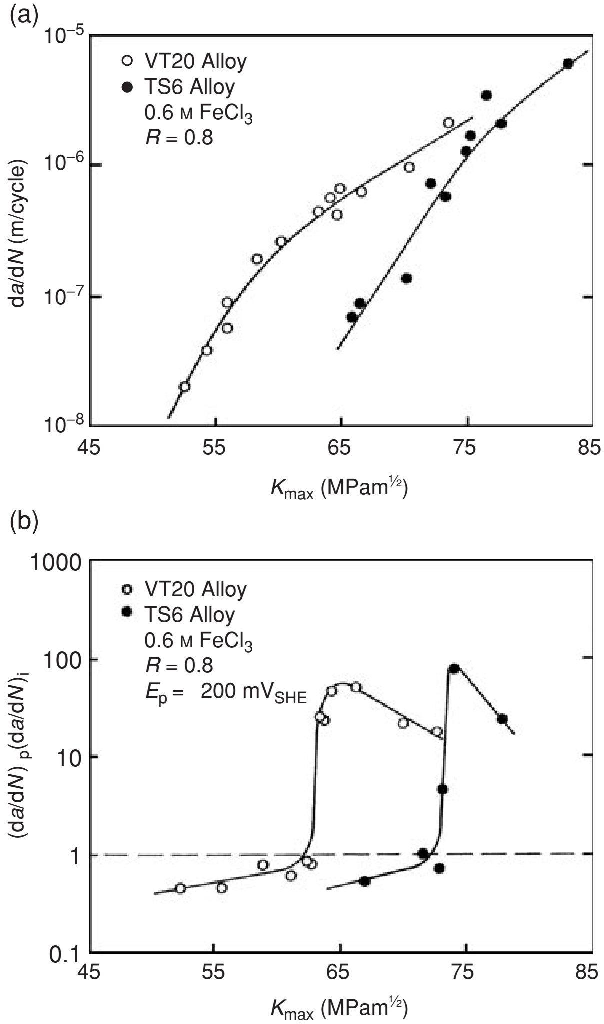

Fig. 6 Dependence of: (a) the corrosion FCG rate, and (b) the $(da/dN)_p/(da/dN)_i$ ratio on K_{max} for titanium alloys in 0.6 M $FeCl_3$ solution.

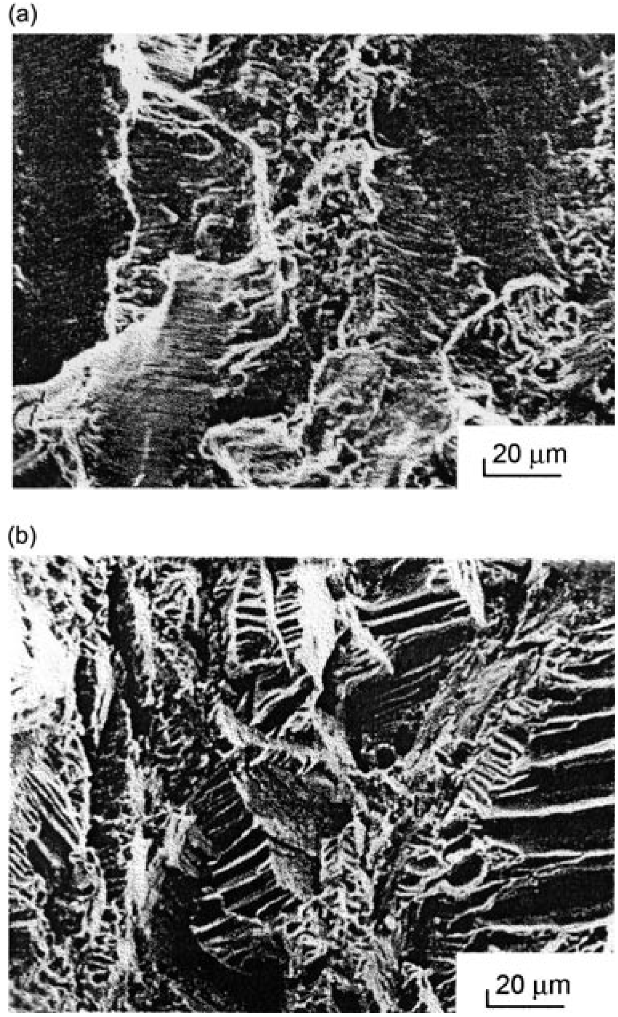

Fig. 7 SEM fractographs of VT20 alloy tested in 0.6 M $FeCl_3$ solution at: (a) OCP of $+750\,mV_{SHE}$ and (b) at cathodic potential of $-200\,mV_{SHE}$; $K_{max} = 68\,MPa\sqrt{m}$, R = 0.5, $(da/dN)_i = 8.5 \times 10^{-7}\,m\,cycle^{-1}$, and $(da/dN)_p = (5\text{–}9) \times 10^{-5}\,m\,cycle^{-1}$. The overall direction of crack propagation was from top to bottom of the fractographs approximately.

When the concentration of Na_2HPO_4 was increased from 0.02 to 0.05 M, the effect of cathodic polarization on crack growth rate also changed. Whereas cathodic polarization accelerated crack growth in 0.02 M Na_2HPO_4 solution at K_{max} higher than a certain critical value (e.g. $10.4\,MPa\sqrt{m}$ at $-1800\,mV_{SHE}$ and $R = 0.5$), in 0.05 M Na_2HPO_4 solution the same cathodic polarization retarded crack growth and this effect was not dependent on K_{max}.

Examination of fracture surfaces of VMD10 alloy revealed no difference between fracture areas corresponding to crack growth under open circuit and to accelerated crack growth under cathodic polarization when K_{max} exceeded the critical values. The TG cracking was revealed on all specimens tested in (0.1–1) M NaOH, 0.02 M Na_2HPO_4, and (0.5–2) M H_2CrO_4 solutions in which the non-single mode effect of cathodic polarization on corrosion FCG rates was observed.

DISCUSSION

Preamble

In aqueous environments, anodic dissolution of metal at the crack tip and the entrance of hydrogen derived from the environment[46] into the crack-tip plastic zone occur simultaneously during corrosion fatigue crack propagation. As a result, both SAD and HIC are involved in the acceleration of crack growth in comparison with crack growth in an inert atmosphere such as, for example, dry argon or high vacuum. [Note: Crack growth due to

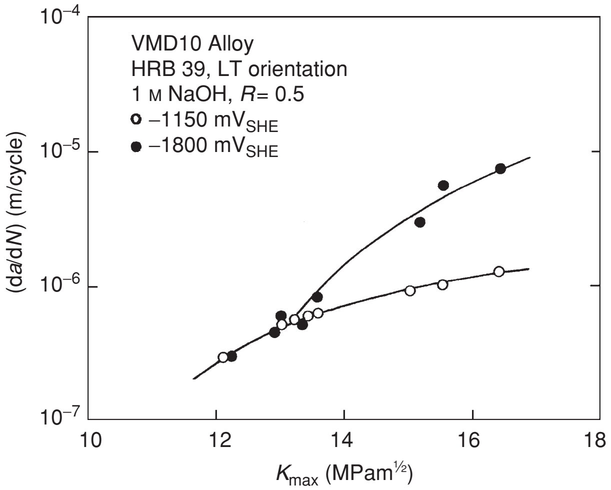

Fig. 8 Corrosion FCG rates for VMD10 alloy in 1 M NaOH solution at OCP ($-1150\,mV_{SHE}$) and cathodic potential ($-1800\,mV_{SHE}$).

internal hydrogen occurring in a material (such as, e.g. titanium alloy) when testing in an inert atmosphere was not included in this analysis.] Depending on mechanical (stress level, mode of loading, cyclic frequency, stress ratio, waveform), metallurgical (chemical composition and microstructure of material, grain boundary composition, surface condition, specimen orientation) and environmental (composition and concentration of solution, pH, potential, dissolved oxygen, temperature, flow rate) variables, one of the two mechanisms plays a dominant role in crack propagation. It should also be noted that the intensity of anodic and/or cathodic reactions in the vicinity of the crack tip, the hydrogen-dependent properties of material and the hydrogen content in the crack-tip zone also determine the controlling corrosion FCG mechanism. Generally, anodic polarization increases metal dissolution and decreases hydrogen generation, whereas cathodic polarization decreases metal dissolution and increases hydrogen generation. This statement identifies the controlled-potential technique[32–34] as a technique for distinguishing between HIC and SAD as possible mechanisms of corrosion FCG by studying the effect of applied potential on the corrosion FCG rate.[5–14,16,35] According to this technique, if an impressed cathodic current increases crack growth rate and an anodic current decreases it, the crack growth mechanism would be HIC. Similarly, the mechanism is considered SAD if crack growth rate decreases with the application of cathodic current and increases with the application of anodic current.[32–34] However, the non-single mode effect of cathodic polarization on corrosion FCG rates which was observed in this study indicated that, depending on the K_{max} value, a given applied cathodic potential either increased by several times or decreased (or had no influence on) crack growth rates comparable to those at OCP could not be explained from the standpoint of HIC or SAD only. On the other hand, the discovered effect provided a reason to conceive a possibility for simultaneous realization of both mechanisms in the crack propagation. Here, the role of each individual mechanism was determined by the K_{max} value (in addition to other variables) that was either lower or higher than a certain critical value characteristic for a given material–solution combination. The interpretation of the results obtained in this study and the explanation of the non-single mode effect of cathodic polarization on corrosion FCG rates were carried out using views and theories developed elsewhere for understanding crack growth mechanism(s) in metals preliminary charged by hydrogen or exposed to hydrogen-producing environments such as aqueous solutions or gaseous hydrogen.

HIC during corrosion fatigue crack propagation

Notion of K_{HIC}, ΔK_{HIC} and $(da/dN)_{cr}$ parameters

The non-single mode effect of cathodic polarization on corrosion FCG rates and the well-defined threshold character of critical values of K_{max} which were obtained under cathodic polarization could not be explained by a change in the electrochemical reactions within the growing crack – although this change was due to the change of potential – because, conceptually, these reactions do not depend on the stress intensity factor. In some metal–solution systems, even for highly conductive solutions, the local chemistry within the crack may not be the same as the chemistry of the bulk solution.[19,47–49] However, there was no reason to relate this difference as well as the possible difference between the applied (external) potential and the potential at the crack tip[49,50] with critical values of K_{max}. These differences could not explain why only very small increases beyond the critical K_{max} reversed the effect of cathodic polarization and accelerated crack growth instead of retarding it. On the other hand, an increased generation of hydrogen under cathodic polarization and its entrance into metal occurred on the external surface of the specimen[46] and within the growing crack.[49,51] In the case being considered, FCG provided local breakdown of the passive film at the crack-tip surface and produced a bare (film-free) metal surface during the opening part of each stress cycle,[17,18,20] for example, by a mechanism of alternating sliding-off at the crack front.[41,52] This promoted entry of hydrogen into the metal which was transported under the driving force of the stress gradient to the region of highest dilation ahead of the crack where some form of hydrogen embrittlement could take place, presumably, by a reduction in the cohesive strength of the lattice.[53,54] The above arguments as well as the fact that susceptibility to HIC under cathodic polarization is inherent to high-strength steels,[55,56] titanium alloys,[57–59] and magnesium alloys,[26,30] suggested that the accelerated corrosion FCG in the materials tested at cathodic potentials was caused by hydrogen charging and HIC. (Others[5–14,16,18] have explained the deleterious effect of cathodic polarization on corrosion FCG rates in a similar way.) However, it is known that stress intensity factor has only a small influence on the rate of crack-tip hydrogen geneation[60] and there is no sharp dependence of the kinetics of hydrogen penetration into metals either on stress or stress intensity factor.[61–63] Moreover, in this study, the same cathodic potential which increased crack growth rates when K_{max} was higher than a certain critical value, decreased (or had no influence on) crack growth rates when K_{max} was lower than the critical value.

Most theories of HIC are based upon the original proposal of Troiano *et al.*[64] about stress-induced diffusion of hydrogen to the point of maximum triaxial stress in the metal matrix ahead of a plastically strained notch or crack. According to this proposal, HIC occurs when sufficient hydrogen has concentrated at this point so that a critical combination of stress state and hydrogen concentration is attained. [Note: More recently, Doig and Jones[65] have assumed that for HIC the critical concentration of hydrogen should be achieved over an unspecified distance ahead of the crack determined by the microstructure. Akhurst,[66] by analogy with brittle modes of unstable fracture (no hydrogen) in steels,[67] has assumed that the critical stress should be exceeded over a characteristic distance ahead of the crack tip for HIC to occur.] Views on the mechanisms of hydrogen-induced FCG in high-strength steels[10,68–70] and titanium alloys[14,71,72] and HIC in magnesium alloys[26,28] incorporated Troiano's proposal too. Moreover, as Oriani and Josephic[73] and Gerberich *et al.*[72,74] proposed, there exists a threshold stress intensity factor (K_{TH}) of which achievement is required to cause a stable crack growth in a metal exposed to hydrogen atmosphere. According to Gerberich and Chen,[74] the stress intensity factor equals K_{TH} when the equilibrium concentration of hydrogen (C_{eq}) ahead of the crack tip achieves the critical level of concentration (C_{cr}) which is required to cause HIC. However, when the stress intensity factor is lower than K_{TH} and $C_{eq} < C_{cr}$, equilibrium does not achieve the critical event and no cracking occurs at long time. More recently, Gerberich *et al.*[72] proposed that by analogy with K_{TH}, there exists a fatigue threshold ($K_n = K_{TH}/\alpha_2$ where $\alpha_2 > 1$) above which hydrogen-induced FCG can occur. This was confirmed experimentally by Suresh and Ritchie[54] who studied the role of (138 kPa) hydrogen gas on FCG rates in several classes of steels. They observed an acceleration in crack growth rates (up to 20 times) when K_{max} exceeded a certain well-defined critical value (K^T_{max}). The crack growth at $K_{max} > K^T_{max}$ occurred by a predominantly IG mode whereas below K^T_{max}, where d*a*/d*N* did not depend of hydrogen presence, the mode of crack growth was TG. Suresh and Ritchie[54] concluded that the acceleration in crack growth rates was related to the onset of hydrogen-induced fatigue crack propagation. In line with all the above findings, it was reasonable to suggest that K_{max}, above which cathodic polarization accelerated the corrosion FCG in this study, equated the threshold value (similar to K_{TH} for HIC in hydrogen atmosphere,[72–74]) which was required to cause the hydrogen-induced crack growth acceleration during corrosion fatigue. This suggestion is in accord with Fujii and Smith[10] who found that cathodic polarization at $-1000\,mV_{SCE}$ (versus saturated calomel electrode) caused an approximately twofold acceleration in the corrosion FCG of high-strength steel HY-130 in 0.6 M NaCl solution. The crack growth acceleration occurred when the stress intensity range ($\Delta K = K_{max} - K_{min}$) exceeded 30–40 $MPa\sqrt{m}$ at $R = 0.1$ and was attributed to the appearance of HIC.[10]

Taking everything stated above into consideration, it was concluded that the critical values of K_{max} which should be exceeded to cause accelerated corrosion FCG at cathodic potentials due to HIC corresponded to conditions when sufficient hydrogen has concentrated at a certain characteristic distance[65,66] ahead of the crack so that the critical combination of stress state and hydrogen concentration[64] was attained. Therefore, the critical values of K_{max} as well as ΔK were regarded as corresponding to the onset of corrosion FCG according to the HIC mechanism and designated as K_{HIC} and ΔK_{HIC} (by analogy with threshold stress intensities K_{FSCC} and ΔK_{SCC} for SCC under cyclic loading,[41,70,75,76]). The corresponding values of critical crack growth rates were designated as $(da/dN)_{cr}$. It is important to note that the conclusion about the dominant role of HIC in corrosion fatigue crack propagation at $K_{max} > K_{HIC}$ ($\Delta K > \Delta K_{HIC}$) and $da/dN > (da/dN)_{cr}$ was common for high-strength low-alloy steels, near-α and near-β titanium alloys, and the magnesium alloy VMD10, i.e. for materials of different base, composition and microstructure. This is consistent with basic theories of HIC[53,64,65,74,77,78] which were proposed for materials regardless of their base, composition and microstructure.

As mentioned, the accelerating effect of cathodic polarization on crack growth rates at $K_{max} > K_{HIC}$ was more pronounced with higher *R* [Fig. 2(b)]. This testified that the relative contribution of HIC (in comparison with another possible mechanism, e.g. SAD) to corrosion FCG increased with increasing *R*. The contribution of HIC to crack propagation was maximal under constant loading. For example, cathodic polarization at

$-200\,mV_{SHE}$ increased the corrosion FCG rate in TS6 alloy exposed to $0.6\,M$ $FeCl_3$ solution by a factor of 5 at $R = 0.1$, 30 at $R = 0.5$, and 70 at $R = 0.8$. Under static loading, the same cathodic polarization caused the growth of the crack, that was 'motionless' at OCP, at the rate of over $4 \times 10^{-5}\,m\,s^{-1}$ whereas K_{HIC} was lower than K_{ISCC}. In accordance with Troiano's proposal on HIC,[64] the above observation was related to the reversible character of stress-induced diffusion of hydrogen into and out of the crack-tip region as the load is reversed. This means that if under static loading hydrogen atoms intend to concentrate in a certain place in the metal matrix ahead of the crack, under cyclic loading hydrogen atoms would fluctuate due to cyclic moving of triaxial position within a wide range[69,79] expanded with decreasing R. It seemed that the cyclic fluctuation of triaxial position and hydrogen atoms ahead of the fatigue crack made more difficult the appearance of HIC comparable with that under static loading and decreased the relative contribution of HIC to corrosion fatigue crack propagation with decreasing R.

When testing VMD10 alloy, K_{HIC} decreased as the yield strength of the alloy increased. This finding agreed with general trends toward increasing susceptibility of materials to HIC with increasing strength. However, the strength was not the primary criteria for susceptibility of magnesium alloys to HIC. For example, IMV7-1 alloy after ageing at 220 °C for 24 h (applied to increase its susceptibility to HIC) was significantly stronger than VMD10 alloy, but the higher strength of IMV7-1 alloy did not result in the acceleration of corrosion FCG under cathodic polarization due to HIC. Cathodic polarization always retarded corrosion FCG in this alloy. That is, susceptibility of IMV7-1 alloy to HIC during corrosion fatigue was less evident than in VMD10 alloy. It seemed that susceptibility to HIC depended on a number of variables rather and not only on the strength of magnesium alloys. Among the variables are the chemical composition and microstructure of the alloy, the solution pH, the ratio of activating/passivating species, the electrode potential, which are responsible for kinetics of crack corrosion processes and the properties of the surface film at the crack tip (including its hydrogen permeability), as well as the K_{max} value were of primary importance. In addition, a strong dependence between K_{HIC} and specimen orientation was observed and it testified strong anisotropy of susceptibility to HIC. In particular, for a SL specimen K_{HIC} was lower than for a LT specimen by a factor of 2.7 when testing VMD10 alloy in $1\,M$ NaOH solution at $-1800\,mV_{SHE}$ and by a factor of 2.9 when testing VMD10 alloy in $2\,M$ H_2CrO_4 solution at $-800\,mV_{SHE}$; where $R = 0.5$. Data on the relationship between specimen orientation of magnesium alloys and their susceptibility to HIC was not found in the literature, but Moccari and Shastry[27] reported a slight effect of specimen orientation on K_{ISCC} of magnesium alloy AZ61 tested in $0.6\,M$ NaCl + $0.1\,M$ K_2CrO_4 solution.

Incubation period for hydrogen-induced corrosion FCG

As mentioned, accelerated crack growth at $K_{max} > K_{HIC}$ did not begin directly after cathodic polarization was applied but only after some delay time (Figs 1 and 5). Delays of 10–150 s were not associated with time required for the penetration of polarization inside the crack because in highly conductive electrolytes, as those used in this study, the change in the crack-tip potential occurs once the potential is applied at the external surface of the specimen. The delays seemed to reflect the kinetics for processes associated with an incubation period interpreted by Troiano[64] as the time required for the critical amount of hydrogen to concentrate in the highly strained region ahead of the crack. Broadly speaking, these delays were associated with an incubation period for the beginning of the corrosion FCG according to HIC mechanism. After the incubation period, hydrogen-induced corrosion fatigue crack grew with steady-state rate $(da/dN)_p$ at a given K_{max} because cathodic polarization provided steady-state generation of hydrogen and, as a result, hydrogen charging of specimens.[50] After cathodic polarization, i.e. the intense surge of hydrogen, was removed and potential was returned to OCP, crack growth rates in steels and VMD10 alloy returned to initial values $(da/dN)_i$. When testing titanium alloys, crack growth rates after cathodic polarization was removed and potential was returned to OCP significantly exceeded the initial value $(da/dN)_i$ for some time. This (residual) effect of cathodic polarization on crack growth rates was due to the greater tendency of titanium and its alloys to the formation of brittle hydrides (particularly under charging conditions) and embrittlement when exposed to a hydrogen-producing environment or gaseous hydrogen.[57,58,71,80–86]

Fractography

The conclusion about the dominant role of HIC in corrosion fatigue crack propagation at $K_{max} > K_{HIC}$ was supported by fractographic observations. The features which were observed on the fracture surfaces produced by accelerated corrosion FCG in high-strength steels and titanium alloys at $K_{max} > K_{HIC}$ were similar to those found for the materials tested under cyclic loading in hydrogen gas or after preliminary hydrogen charging.[55–57,82,87,88] In particular, IG cracking mixed with patches of TG cracking and some evidence of inelastic incremental crack advance occurring along grain boundaries was observed in high-strength steels [Fig. 3(b)]. It is typical for HIC of steels of this type. In general, the hydrogen-induced corrosion FCG in martensitic steels at cathodic potentials occurs intergranularly along prior austenite grain boundaries.[5,12,16,89] Frandsen and Marcus[87] found a good correlation between the acceleration in FCG of high-strength steel HP-9-4-20 tested in (13 kPa) hydrogen gas relative to vacuum and the proportion of IG cracking on the fracture surface. They proposed that prior austenite grain boundaries might be a preferred fracture path for HIC due to stress concentration created by blocked slip bands at the boundary[90] or due to the presence of various impurities which acted as hydrogen recombination poisons.[91] It is now widely accepted that high-strength martensitic steels generally exhibit brittle fracture along prior austenite grain boundaries when stressed in hydrogen gas or hydrogen-producing environments.[78,91,92] The tendency to IG cracking is directly related to the concentration of trace impurities such as P, S, and Sn[78,93,94] which are known to reduce IG cohesion in iron-base alloys.[95] In addition, McMahon *et al.*[78,94] demonstrated that alloying elements such as Mn, Si and Ni promote segregation of P and S at austenite grain boundaries. In steels tested in this study, all those alloying elements and impurities (except for Sn, which was immediately discernable) were present (Table 1). It was suspected that they promoted the IG propagation of hydrogen-induced corrosion fatigue crack in steels at $K_{max} > K_{HIC}$.

Corrosion FCG in titanium alloys under cathodic polarization at $K_{max} > K_{HIC}$ occurred in a more brittle manner than under open circuit. Figure 7(b) shows that many typical flat cleavage facets with characteristic 'river patterns' and fissures perpendicular to the fracture surface and parallel to the main crack growth direction were evident after accelerated corrosion FCG under cathodic polarization. Hack and Leverant[82] and, more recently, Nakasa and Satoh,[57] performed microstructural characterization and fractographic analysis of fracture surfaces of titanium alloys fatigue tested under and after hydrogen charging. Almost complete brittle fracture and numerous cleavage facets on the fracture surface of hydrogen-charged IMI-685 alloy were associated with HIC.[82] The fracture surface of Ti-13V-11Cr-3Al alloy cathodically charged with hydrogen revealed the cleavage fracture with frequent 'river patterns' which corresponded to the decrease in the cohesive strength of cleavage plane of β-phase by the hydrogen solution.[57] The mechanisms which could cause a cleavage of titanium lattice due to the action of hydrogen have been discussed by Meyn,[83] and by Powell and Scully.[96]

TG cracking was revealed on the fracture surfaces of VMD10 alloy after testing in (0.1–1) M NaOH, 0.02 M Na_2HPO_4, and (0.5–2) M H_2CrO_4 solutions in which the non-single mode effect of cathodic polarization on crack growth rates was observed. The morphology of fracture surfaces virtually did not depend on the potential and specimen orientation when K_{max} exceeded K_{HIC}. It seemed likely that this was related to the fact that the acceleration effect of cathodic polarization on corrosion FCG rates at $K_{max} > K_{HIC}$ was less pronounced in VMD10 alloy than in high-strength low-alloy steels and titanium alloys.

SAD during corrosion fatigue crack propagation

The appearance of two fractographic modes of corrosion FCG at $K_{max} > K_{HIC}$ and $K_{max} < K_{HIC}$ suggested that there might be two different mechanisms of crack growth under these conditions. It seemed that, at $K_{max} < K_{HIC}$ the critical combination of stress state and hydrogen concentration has not been achieved ahead of the growing crack. Because of this, although the applied cathodic potential was far below the hydrogen equilibrium potential[43] and the hydrogen content ahead of the crack tip could be higher than under open circuit conditions, the accelerated crack growth due to HIC did not occur. On the other hand, cathodic polarization retarded corrosion FCG in high-strength steels and titanium alloys at $K_{max} < K_{HIC}$. Although the results were obtained in passivating solutions in which SCC due to SAD was suppressed as evident in the increase of K_{ISCC}, it is reasonable to suggest that a SAD-based mechanism played a dominant role in the corrosion fatigue crack propagation at $K_{max} < K_{HIC}$. This suggestion is supported by the visible displacement of OCP (relative to the OCP of an unloaded specimen) during corrosion FCG as well as its periodical fluctuation with the cyclic loading. For example, when testing VT20 alloy in 0.5 M $H_2CrO_4 + 0.1$ M NaCl solution, OCP was ennobled slightly at $da/dN < 1.3 \times 10^{-7}$ m cycle^{-1} but OCP was displaced cathodically at higher da/dN such that its mean displacement reached 350 mV at $da/dN = 10^{-5}$ m cycle^{-1}. In addition, during the opening part of each cycle, OCP became more negative due to the formation of bare (film-free) metal surface at the crack tip. After reaching the maximum load of stress cycle, OCP returned – due to repassivation – to its previous value displaced relative to the OCP of an unloaded specimen. For VT20 alloy tested in 0.5 M $H_2CrO_4 + 0.1$ M NaCl solution, the maximum cyclic fluctuation of OCP during a fatigue cycle was about 30 mV for higher crack growth rates.

Independently of K_{max} value, SAD was also the main mechanism of corrosion FCG in titanium alloys when tested in 0.5 M NaCl, (0.1–1) M NaOH with and without NaCl, (0.1–1) M KOH with and without NaCl, and (0.5–2) M H_2CrO_4. In these solutions, cathodic polarization retarded and even stopped corrosion FCG (Fig. 4), whereas anodic polarization accelerated crack growth.

It seemed that the crack growth mechanism in VMD10 alloy tested in (0.1–1) M NaOH, 0.02 M Na_2HPO_4, and (0.5–2) M H_2CrO_4 solutions at $K_{max} < K_{HIC}$ was not electrochemical or the role of electrochemical processes was not significant in the crack propagation because da/dN data obtained in the solutions were like that of da/dN data in air and cathodic polarization had no influence on crack growth rates for a time up to 1500 s. [Note: da/dN data obtained in air cannot be used as a reference to compare the effect of aqueous solutions on the FCG behaviour of magnesium alloys because laboratory air is an aggressive environment for magnesium and its alloys in comparison with an inert environment such as vacuum.[45]] Because cathodic polarization retarded corrosion FCG in IMV7-1 alloy whereas anodic polarization accelerated it, and the polarization effect did not depend on solution composition, applied potential and K_{max} value, it is suggested that SAD played a dominant role in the corrosion fatigue crack propagation of this alloy.

In this study, the conclusion that one of the mechanisms played a dominant role in the corrosion fatigue crack propagation does not deny the possibility that other, less evident, mechanisms were involved. For example, although cathodic polarization retarded and even stopped corrosion FCG in VT20 alloy tested in 0.5 M NaCl solution (and this fact caused difficulty in the acceptance of HIC mechanism), many features of the fracture surfaces produced when testing VT20 alloy in this solution at OCP were similar to those found for VT20 alloy tested in 0.6 M $FeCl_3$ solution at $K_{max} > K_{HIC}$ when HIC was the main mechanism of corrosion FCG.[38] It seemed that the similarity between the features of the fracture surfaces was due to coincidence of crack growth rates (3×10^{-5} m cycle^{-1} in 0.5 M NaCl and 5.3×10^{-5} m cycle^{-1} in 0.6 M $FeCl_3$) rather than to a coincidence of crack growth mechanisms. As demonstrated by Lynch,[97] the features of fracture surfaces produced under very different conditions can be quite similar. The observation that the features of fracture surfaces depend on the level of crack growth rates rather than on the crack growth mechanism suggests that fracture surfaces should not be identified with a crack growth mechanism without additional evidence, including the effect of controlled (applied) potential on the crack growth rate.

CONCLUSIONS

1 A new approach was proposed to identify the role of HIC and SAD in corrosion fatigue crack propagation. The approach was intended for short-term stimulation of electrochemical reactions within a growing crack. For this purpose, anodic or cathodic potential was applied to the specimen for a time period in which it was possible to register the change in the crack growth rate corresponding to OCP and to measure the crack growth rate under polarization. Due to the high resolution of the technique used to measure the crack extension, the time rarely exceeded 300 s.

2 The approach made possible the observation of a non-single mode effect of cathodic polarization on corrosion FCG rates. An important point was that cathodic polarization accelerated crack growth when K_{max} and the corresponding crack growth rate exceeded certain well-defined critical values characteristic for a given material-solution combination. When K_{max} and da/dN were lower than the critical values, the same cathodic polarization, with all other conditions (specimen, solution, pH, temperature, frequency, R, etc.) being equal, retarded or had no obvious influence on crack growth. The observed effect was common for martensitic steels, near-α and near-β titanium alloys, and the magnesium alloy VMD10.

3 The features that were observed on the fracture surfaces produced by accelerated corrosion FCG in high-strength steels and titanium alloys at cathodic potentials were similar to those found for the materials fatigue tested in gaseous hydrogen or after preliminary hydrogen charging.

4 Results suggested that the acceleration in corrosion FCG at cathodic potentials was due to HIC. Therefore, the critical values of K_{max} as well as ΔK and da/dN were regarded as corresponding to the onset of corrosion FCG according to the HIC mechanism and designated as K_{HIC}, ΔK_{HIC} and $(da/dN)_{cr}$. HIC was the main mechanism of corrosion FCG at $K_{max} > K_{HIC}$ ($\Delta K > \Delta K_{HIC}$) and $da/dN > (da/dN)_{cr}$.

5 SAD-based mechanism played a dominant role in the corrosion fatigue crack propagation at $K_{max} < K_{HIC}$ and $da/dN < (da/dN)_{cr}$. In addition, SAD was the main mechanism of corrosion FCG in titanium alloys and magnesium alloys in most solutions investigated in which cathodic polarization retarded and even stopped corrosion FCG.

6 The relative contribution of HIC (in comparison with another possible mechanism, e.g. SAD) to corrosion fatigue crack propagation increased with increasing R. Under static loading, HIC was more pronounced than under cyclic loading.

Acknowledgements

The author acknowledges helpful discussions of V. V. Lunin (deceased) and V. A. Marichev of the Institute of Physical Chemistry of the Russian Academy of Sciences, and the assistance of L. N. Yuvenalyeva of the Central Materials Research Institute with microscopy. The editorial assistance of H. M. Simpson of York University is also gratefully acknowledged.

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